



Inputs

Paper



extract

Reference repos



Channel Processing

Implementation-Requirement Channel

Code-needed snippets

```
</> ...
```

```
def ...
```

```
import ...
```

Implementation requirements



must keep

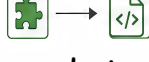
Reference-Evidence Channel

Content evidence



Find code pieces

reuse



adapt

function

class

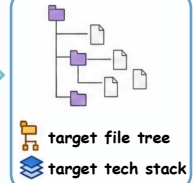
config

script

Structure evidence



current task



Shared Contract

Implementation check



Output

Requirement review

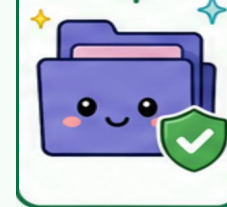


Runtime check



y

Faithful runnable repo



Validation & Repair Loop

- ✓ Paper context kept
- ✓ Missing details filled